

IND320-OLEGR099\D2Dbook\streamlit app\Pages\analysis.py

```
1 import streamlit as st
2 import pandas as pd
3
4
5 @st.cache_data
6 def load_reservoir_data():
7     data = pd.read_csv("../data/reservoirs_processed.csv")
8
9     data = data.rename(columns={
10         "dato_Id": "Date",
11         "fyllingsgrad": "filling_level",
12         "kapasitet_TWh": "Capacity(TWh)",
13         "fylling_TWh": "Filled (TWh)",
14         "fyllingsgrad_forrige_uke": "filling_level_previous_week",
15         "endring_fyllingsgrad": "change_filling_level"
16     })
17
18     data["Date"] = pd.to_datetime(data["Date"])
19
20     return data
21
22
23 reservoir_data = load_reservoir_data()
24
25 st.title("Reservoir data analysis")
26
27 # Columns that can be plotted
28 columns = [
29     "filling_level",
30     "Capacity(TWh)",
31     "Filled (TWh)",
32     "filling_level_previous_week",
33     "change_filling_level"
34 ]
35
36 # Select one column or all columns
37 selected_column = st.selectbox(
38     "Select data to display",
39     ["All"] + columns
40 )
41
42 # Create list of months
43 months = sorted(reservoir_data["Date"].dt.to_period("M").unique())
44
45 # Select months
46 selected_month = st.select_slider(
47     "Select month",
48     options=months,
49     value=months[0]
50 )
51
```

```
52 # Filter data based on selected month
53 filtered_data = reservoir_data[
54     reservoir_data["Date"].dt.to_period("M") == selected_month
55 ]
56
57 # Plot
58 if selected_column == "All":
59     st.line_chart(
60         filtered_data.set_index("Date")[columns],
61         y_label="Value",
62         x_label="Date"
63     )
64 else:
65     st.line_chart(
66         filtered_data.set_index("Date"),
67         y=selected_column,
68         y_label=selected_column,
69         x_label="Date"
70     )
71
72
```